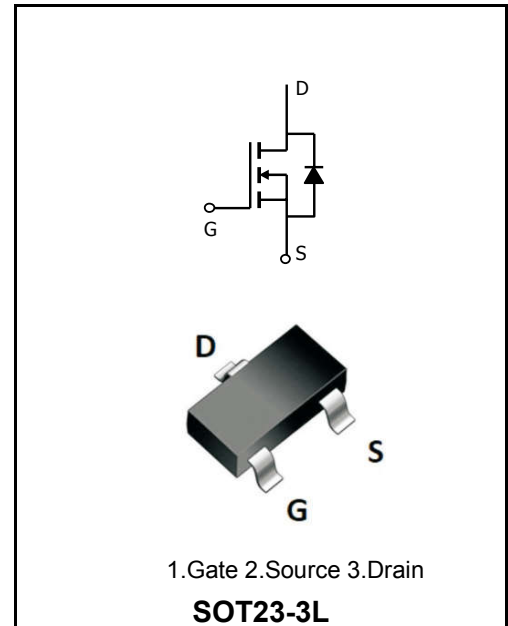


N-Channel Enhancement MOSFET

Features

- $V_{DS} (V) = 30V$
- $I_D = 5.8 A (V_{GS} = 10V)$
- $R_{DS(ON)} < 32m\Omega (V_{GS} = 10V)$
- $R_{DS(ON)} < 36m\Omega (V_{GS} = 4.5V)$
- $R_{DS(ON)} < 52m\Omega (V_{GS} = 2.5V)$



1. Gate 2. Source 3. Drain

SOT23-3L

Marking Code

AO3400

A0H

Absolute Maximum Ratings at $T_A=25^\circ C$

Parameter	Symbol	Rating	Unit
Drain-Source Voltage	V_{DS}	30	V
Gate-Source Voltage	V_{GS}	± 12	V
Continuous Drain Current $T_A=25^\circ C$	I_D	5.8	A
$T_A=70^\circ C$		4.9	
Pulsed Drain Current *	I_{DM}	30	
Power Dissipation $T_A=25^\circ C$	P_D	1.4	W
$T_A=70^\circ C$		1	
Thermal Resistance.Junction- to-Ambient	R_{thJA}	125	$^\circ C/W$
Thermal Resistance.Junction- to-Case	R_{thc}	60	$^\circ C/W$
Junction and Storage Temperature Range	T_J, T_{STG}	-55 to 150	$^\circ C$

* Repetitive rating, pulse width limited by junction temperature.

Electrical Characteristics at Ta25°C

Parameter	Symbol	Testconditions	Min	Typ	Max	Unit
Drain-Source Breakdown Voltage	V _{DSS}	I _D =250 μA, V _{GS} =0V	30			V
Zero Gate Voltage Drain Current	I _{DSS}	V _{DS} =24V, V _{GS} =0V			1	μA
		V _{DS} =24V, V _{GS} =0V, T _J =55°C			5	
Gate-Body leakage current	I _{GSS}	V _{DS} =0V, V _{GS} =±12V			±100	nA
Gate Threshold Voltage	V _{GS(th)}	V _{DS} =V _{GS} I _D =250 μA	0.7	1.1	1.4	V
Static Drain-Source On-Resistance	R _{DS(on)}	V _{GS} =10V, I _D =5.8A		29	32	mΩ
		V _{GS} =10V, I _D =5.8A T _J =125°C			39	
		V _{GS} =4.5V, I _D =5A		32	36	mΩ
		V _{GS} =2.5V, I _D =4A		40	52	mΩ
On state drain current	I _{D(on)}	V _{GS} =4.5V, V _{DS} =5V	30			A
Forward Transconductance	g _{FS}	V _{DS} =5V, I _D =5A	10	15		S
Input Capacitance	C _{iss}	V _{GS} =0V, V _{DS} =15V, f=1MHz		823	1050	pF
Output Capacitance	C _{oss}			99		pF
Reverse Transfer Capacitance	C _{rss}			77		pF
Gate resistance	R _g	V _{GS} =0V, V _{DS} =0V, f=1MHz		1.4	3.6	Ω
Total Gate Charge	Q _g	V _{GS} =4.5V, V _{DS} =15V, I _D =5.8A		9.7	12	nC
Gate Source Charge	Q _{gs}			1.6		nC
Gate Drain Charge	Q _{gd}			3.1		nC
Turn-On DelayTime	t _{D(on)}	V _{GS} =10V, V _{DS} =15V, R _L =2.7 Ω, R _{GEN} =3 Ω		3.3	5	ns
Turn-On Rise Time	t _r			4.8	7	ns
Turn-Off DelayTime	t _{D(off)}			26.3	40	ns
Turn-Off Fall Time	t _f			4.1	6	ns
Body Diode Reverse Recovery Time	t _{rr}	I _F =5A, dI/dt=100A/μs		16	20	ns
Body Diode Reverse Recovery Charge	Q _{rr}	I _F =5A, dI/dt=100A/μs		8.9	12	nC
Maximum Body-Diode Continuous Current	I _S				2.5	A
Diode Forward Voltage	V _{SD}	I _S =1A, V _{GS} =0V		0.71	1	V

Ratings and Characteristic Curves

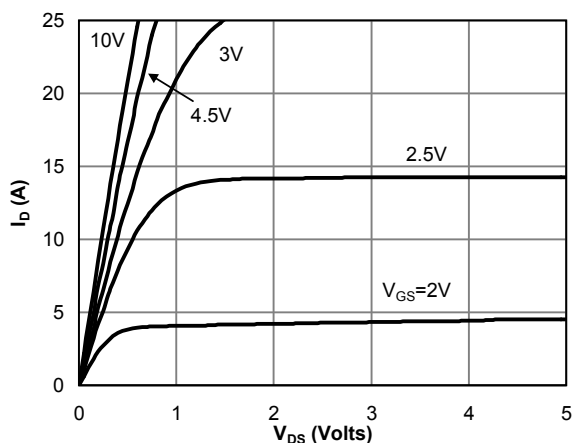


Fig 1: On-Region Characteristics

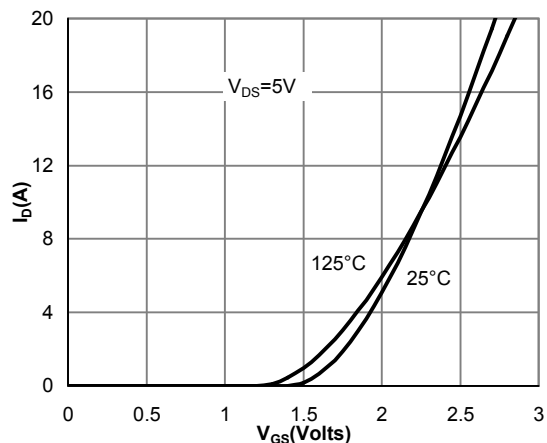


Figure 2: Transfer Characteristics

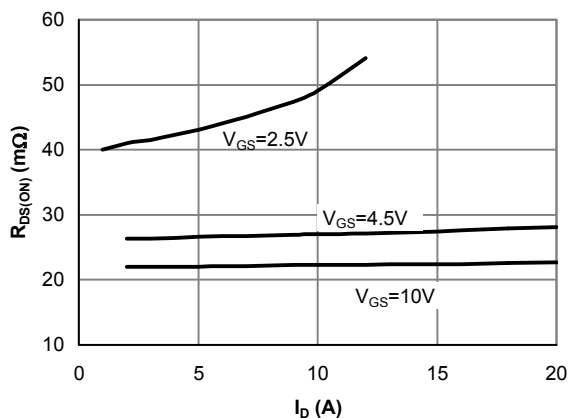


Figure 3: On-Resistance vs. Drain Current and Gate Voltage

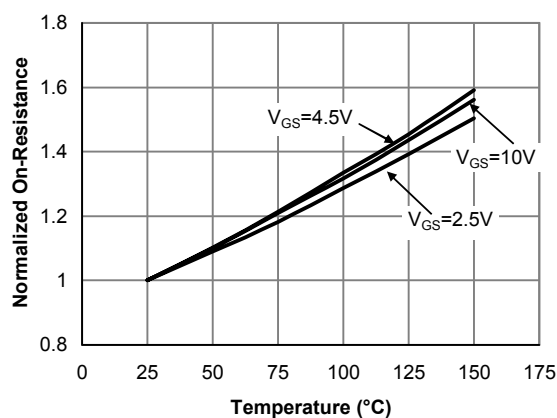


Figure 4: On-Resistance vs. Junction Temperature

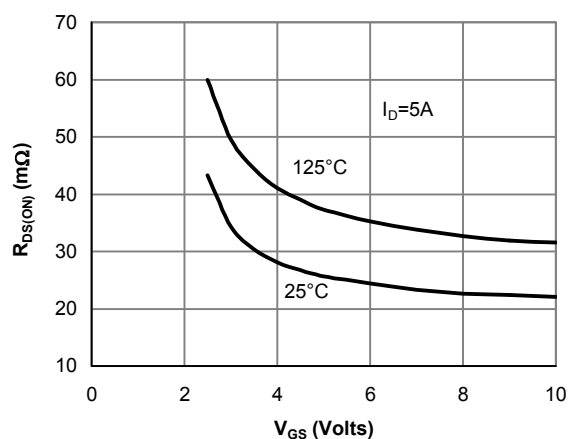


Figure 5: On-Resistance vs. Gate-Source Voltage

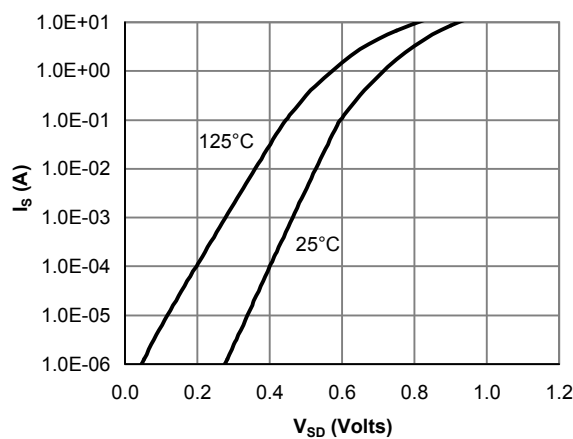


Figure 6: Body-Diode Characteristics

Ratings and Characteristic Curves

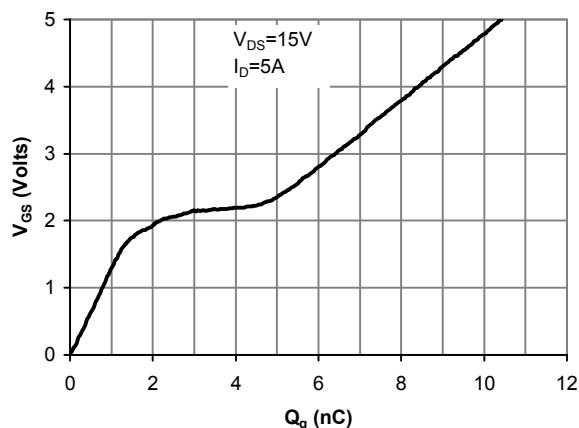


Figure 7: Gate-Charge Characteristics

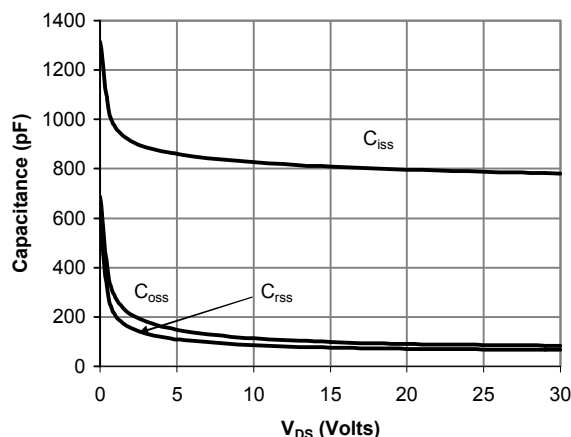


Figure 8: Capacitance Characteristics

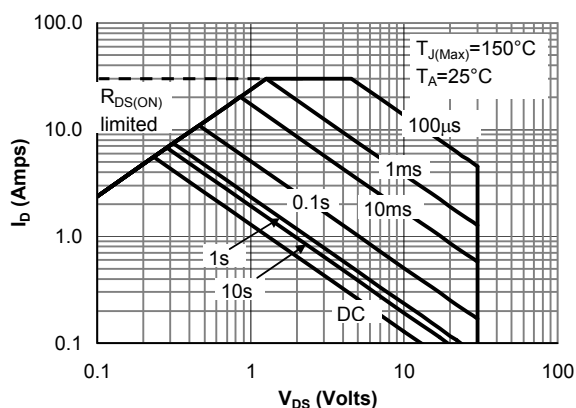


Figure 9: Maximum Forward Biased Safe Operating Area (Note E)

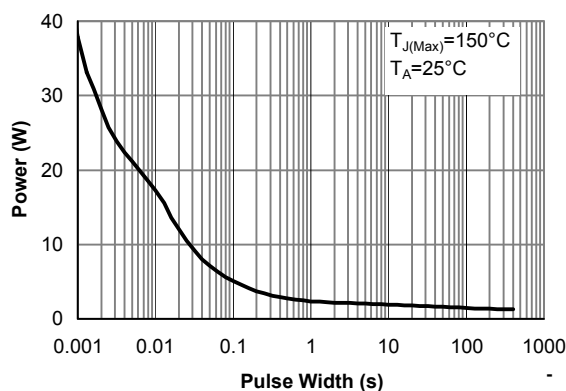


Figure 10: Single Pulse Power Rating Junction-to-Ambient (Note E)

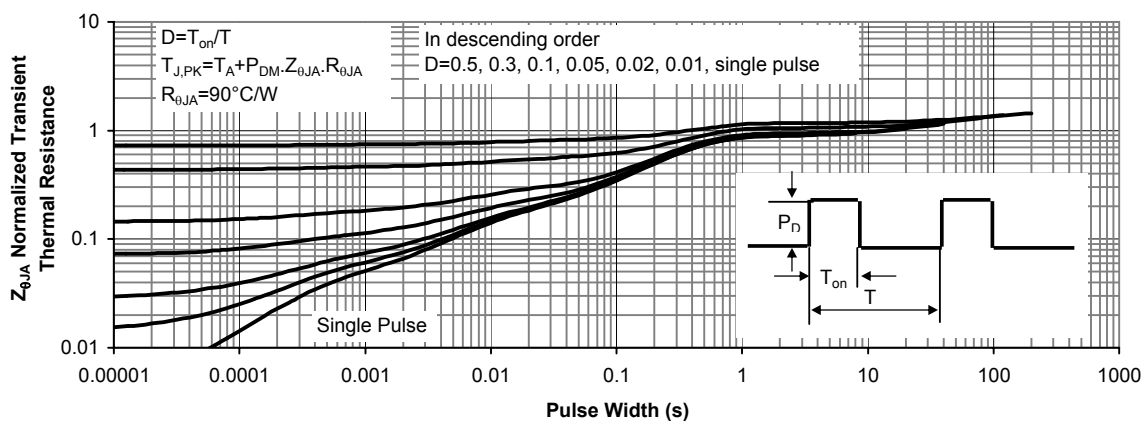


Figure 11: Normalized Maximum Transient Thermal Impedance

Ordering information

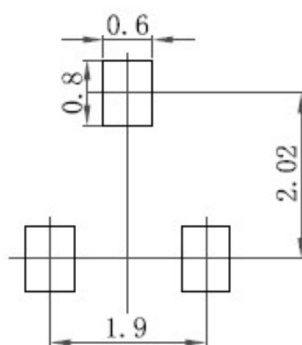
Package	Packing Description	Base Quantity	Packing Quantity
SOT23-3L	Tape/Reel, 7" reel	3000pcs/Reel	24000PCS/Box 120000PCS/Carton

Package Dimensions

SOT23-3L

Dim.	Millimeter (mm)		mil	
	Min.	Max.	Min.	Max.
A	1.05	1.25	41	49.2
A1	0.10		3.93	
A2	1.05	1.15	41	45
b	0.30	0.50	12	20
c	0.10	0.20	3.93	7.9
D	2.82	3.02	111	119
E	1.50	1.70	59	67
E1	2.65	2.95	104	116
e	0.95		37.4	
e1	1.80	2.00	71	78
L	0.30	0.066	12	26
Θ	8°			

The recommended mounting pad size



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